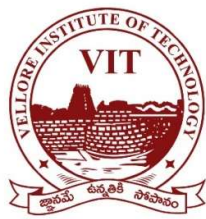


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**VIT-AP
UNIVERSITY**

Fundamental of Electrical and Electronics Engineering

Lecture Notes

School of Electronics Engineering

2020-21

MODULE 4

Operational Amplifier

S.No	Operational Amplifier	Faculty
<u>1</u>	Ideal Op-amp, Differential amplifier: differential and common mode operation, common-mode rejection ratio (CMRR)	Dr.Khairnar Vikas Vishnu
<u>2</u>	Practical op-amp circuits: inverting amplifier, non - inverting amplifier	Dr. T.V Mahendra
<u>3</u>	weighted, summer, integrator, differentiator, Large signal operation of op-amps	Dr. Amarjit Roy
<u>4</u>	Exercise Problems (4 to 6)	Dr. Praveen Kumar
<u>5</u>	MCQ (4 to 6)	Dr. Arvind Kumar Prajapati

LECTURE 1

❖ Introduction

- The term operational amplifier was introduced in 1947 by John Ragazzini and his colleagues, in their work on analog computers.
- The op-amp is an electronic unit that behaves like a voltage-controlled voltage source. It can also be used in making a voltage- or current-controlled current source.
- An op-amp can sum signals, amplify a signal, integrate it, or differentiate it. The ability of the op-amp to perform these mathematical operations is the reason it is called an operational amplifier.

❖ Operational Amplifiers (op-amp)

- An operational amplifier is designed so that it performs some mathematical operations when external components, such as resistors and capacitors, are connected to its terminals. Thus,

An op amp is an active circuit element designed to perform mathematical operations of addition, subtraction, multiplication, division, differentiation, and integration.

- Figure 4.1 shows a typical op-amp package and pin diagram.

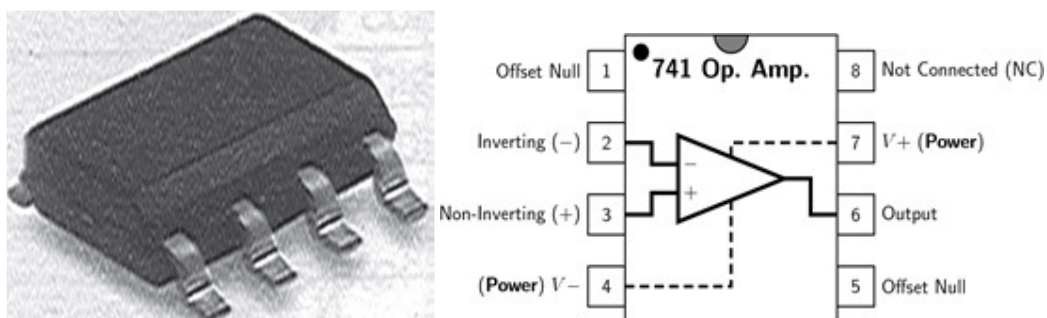


Figure 4.1 A typical operational amplifier and pin diagram

- Pin 8 is unused, and pins 1 and 5 are offset null.

- Pin 2: inverting input, pin 3: noninverting input, pin 6: output, pin 7: positive power supply, pin 4: negative power supply
- The circuit symbol for the op-amp is shown in Figure 4.2. The op-amp has two inputs and one output.

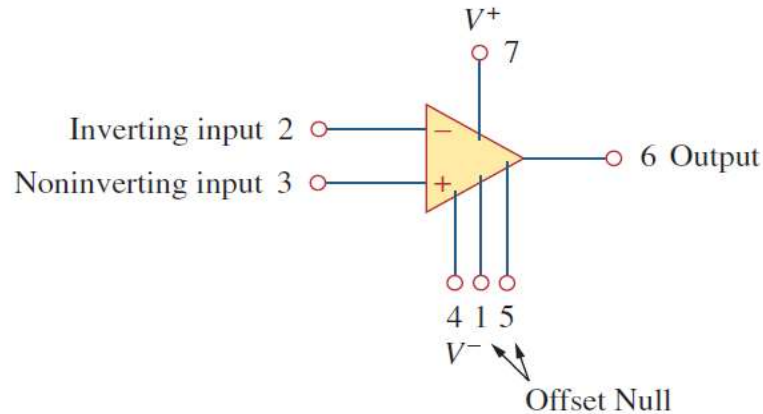


Figure 4.2 Circuit symbol

- The inputs are marked with minus (-) and plus (+) to specify inverting and noninverting inputs, respectively. An input applied to the noninverting terminal will appear with the same polarity at the output, while an input applied to the inverting terminal will appear inverted at the output.
- As an active element, the op amp must be powered by a voltage supply as typically shown in Figure 4.3.

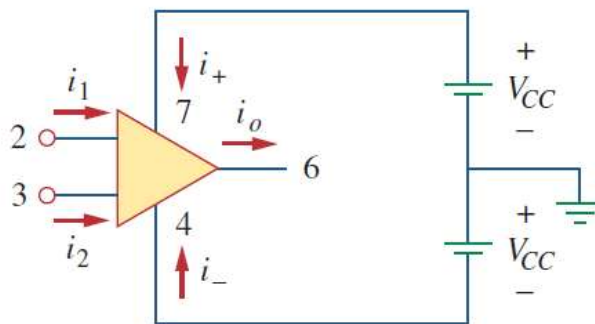


Figure 4.3 Powering the op-amp

- By KCL, $i_0 = i_1 + i_2 + i_+ + i_-$
- The equivalent circuit model of an op-amp is shown in Figure 4.4. The output section consists of a voltage-controlled source in series with the output resistance R_0 .

- It is evident from the Figure 4.4 that the input resistance R_i is the Thevenin equivalent resistance seen at the input terminals, while the output resistance R_o is the Thevenin equivalent resistance seen at the output. The differential input voltage v_d is given by,

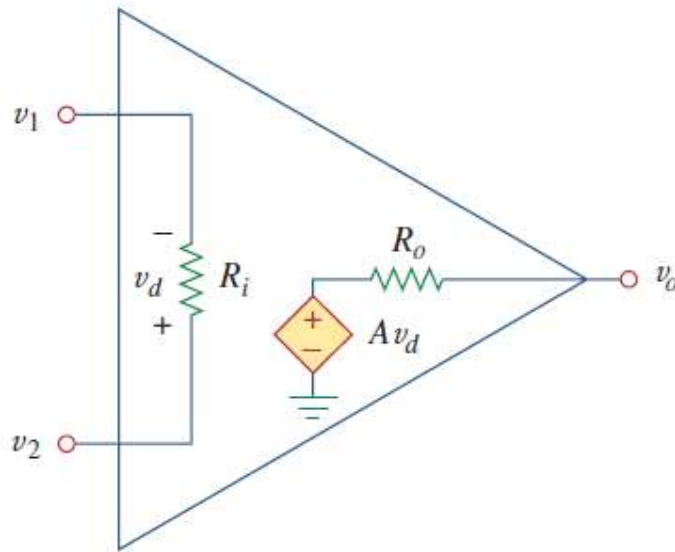


Figure 4.4 The equivalent circuit of the nonideal op-amp

$$v_d = v_2 - v_1$$

- Where v_1 is the voltage between the inverting terminal and ground and v_2 is the voltage between the noninverting terminal and ground. The op-amp senses the difference between the two inputs, multiplies it by the gain A , and causes the resulting voltage to appear at the output. Thus, the output v_o is given by

$$v_o = Av_d = A(v_2 - v_1)$$

- A is called the open-loop voltage gain because it is the gain of the op-amp without any external feedback from output to input

$$A \text{ (dB)} = 20\log_{10} A$$

- Table 4.1 shows typical values of the voltage gain A , input resistance R_i , output resistance R_o , and supply voltage V_{cc}

Table 4.1 Typical ranges for op-amp parameters

Parameter	Typical range	Ideal values
Open-loop gain, A	10^5 to 10^8	∞
Input resistance, R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance, R_o	10 to 100Ω	0Ω
Supply voltage, V_{CC}	5 to 24 V	

- The concept of feedback is crucial to our understanding of op amp circuits. A negative feedback is achieved when the output is fed back to the inverting terminal of the op amp. As Example 5.1 shows,
- When there is a feedback path from output to input, the ratio of the output voltage to the input voltage is called the closed-loop gain. As a result of the negative feedback, it can be shown that the closed-loop gain is almost insensitive to the open-loop gain A of the op-amp. For this reason, op- amps are used in circuits with feedback paths.
- A practical limitation of the op-amp is that the magnitude of its output voltage cannot exceed $|V_{CC}|$. In other words, the output voltage is dependent on and is limited by the power supply voltage.
- Figure 4.5 illustrates that the op amp can operate in three modes, depending on the differential input voltage v_d .

1. Positive saturation, $v_o = V_{CC}$.
2. Linear region, $-V_{CC} \leq v_o = Av_d \leq V_{CC}$.
3. Negative saturation, $v_o = -V_{CC}$.

- The output voltage is restricted by

$$-V_{CC} \leq v_o \leq V_{CC}$$

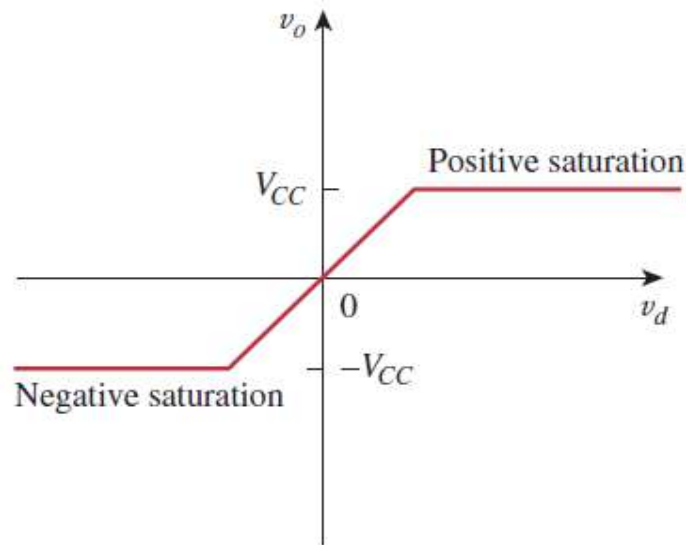


Figure 4.5 op-amp output voltage v_o as a function of the differential input voltage v_d

❖ **Example**

- 1) A 741 op-amp has an open-loop voltage gain of 2×10^5 , input resistance of $2 \text{ M}\Omega$, and output resistance of 50Ω . The op-amp is used in the circuit of Fig. 4.6(a). Find the closed-loop gain v_o/v_s . Determine current i when $v_s = 2 \text{ V}$.

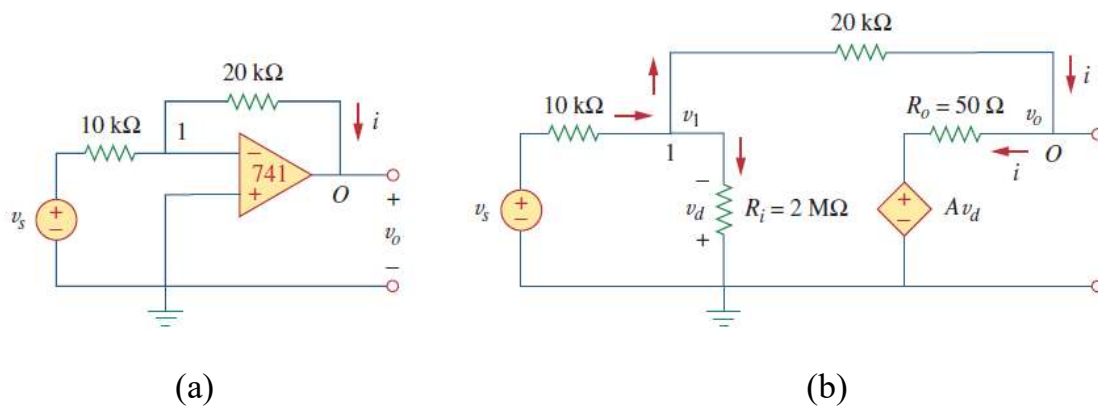


Figure 4.6 (a) original circuit (b) the equivalent circuit

- 2) The open-loop gain of an op-amp is 100,000. Calculate the output voltage when there are inputs of $+10 \mu\text{V}$ on the inverting terminal and $+20 \mu\text{V}$ on the noninverting terminal
- 3) Determine the output voltage when $-20 \mu\text{V}$ is applied to the inverting terminal of an op-amp and $+30 \mu\text{V}$ to its noninverting terminal. Assume that the op-amp has an open-loop gain of 200,000.

LECTURE 2

❖ Ideal op-amp

- An op-amp is ideal if it has the following characteristics
 - a) Infinite voltage gain A
 - b) Infinite input resistance R_i so that almost any signal source can drive it and there is no loading of the preceding stage
 - c) Zero output resistance R_o so that the output can drive an infinite number of other devices
 - d) Zero output voltage when input voltage is zero
 - e) Infinite bandwidth so that any frequency signal from 0 to ∞ Hz can be amplified without attenuation
 - f) Infinite common-mode rejection ratio (CMRR) so that the output common-mode noise voltage is zero
 - g) Infinite slew rate so that output voltage changes occur simultaneously with input voltage changes

- For circuit analysis, the ideal op-amp is illustrated in Figure 4.7, which is derived from the nonideal model in Figure 4.4.

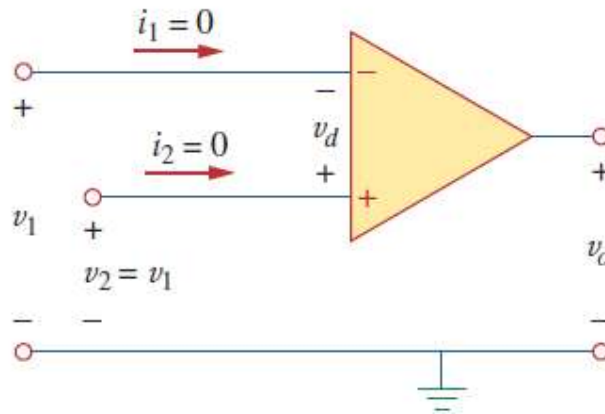


Figure 4.7 Ideal op-amp model

- Two important properties of the ideal op-amp are
 - a) The currents into both input terminals are zero. This is due to infinite input resistance. An infinite resistance between the input terminals implies that an open circuit exists there and current cannot enter the op-amp. But the output current is not necessarily zero.

$$i_1 = 0, \quad i_2 = 0$$

- b) The voltage across the input terminals is equal to zero. Thus, an ideal op-amp has zero current into its two terminals and the voltage between the two input terminals is equal to zero.

$$v_d = v_2 - v_1 = 0$$

$$v_1 = v_2$$

❖ Differential Amplifier

- Differential (or difference) amplifiers are used in various applications where there is a need to amplify the difference between the two input signals.

“A differential amplifier is a device that amplifies the difference between two inputs but rejects any signals common to the two inputs”

- Generally, the differential amplifiers are used in instrumentation and industrial applications to amplify differences between two input signals, such as the outputs of the Wheatstone bridge circuits.
- Differential amplifiers are preferred in these applications because they are better able to reject common-mode (noise) voltages than single-input circuits such as inverting and noninverting amplifiers. They also present a balanced input impedance.
- Consider the differential amplifier circuit shown in Figure 4.8. It can be noted that zero current enters the op-amp terminals.

Apply KCL at node a,

$$\frac{v_1 - v_a}{R_1} = \frac{v_a - v_o}{R_2}$$

or

$$v_o = \left(\frac{R_2}{R_1} + 1 \right) v_a - \frac{R_2}{R_1} v_1$$

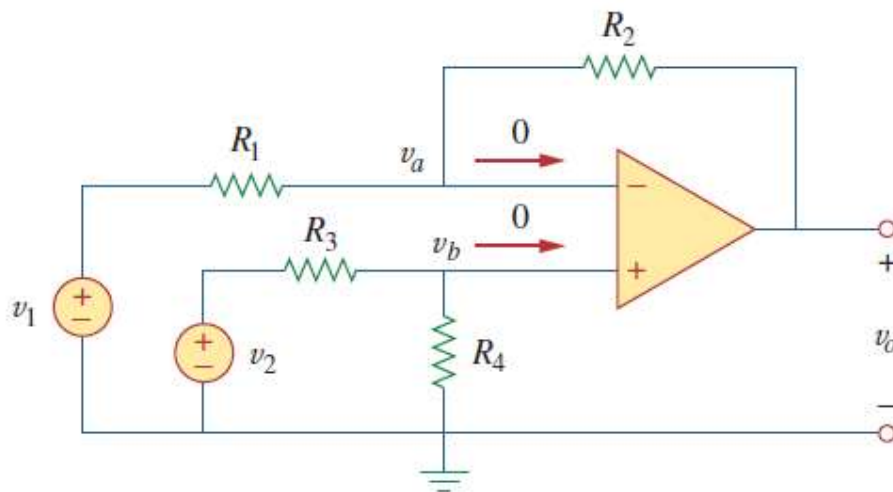


Figure 4.8 Differential amplifier

Apply KCL at node b,

$$\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4}$$

or

$$v_b = \frac{R_4}{R_3 + R_4} v_2$$

But

$$v_a = v_b$$

$$v_0 = \left(\frac{R_2}{R_1} + 1 \right) \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} v_1$$

$$v_0 = \frac{R_2 \left(1 + R_1/R_2 \right)}{R_1 \left(1 + R_3/R_4 \right)} v_2 - \frac{R_2}{R_1} v_1$$

Since a difference amplifier must reject a signal common to the two inputs, the amplifier must have the property that $v_0 = 0$ and $v_1 = v_2$

This property exists when

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Thus, when the op-amp circuit is a difference amplifier, we can have

$$v_0 = \frac{R_2}{R_1} (v_2 - v_1)$$

If $R_2 = R_1$ and $R_3 = R_4$, the differential amplifier becomes a subtractor, with the output

$$v_0 = v_2 - v_1$$

❖ Differential Amplifier

A **differential amplifier** (also known as a **difference amplifier** or **op-amp subtractor**) is a type of electronic amplifier that amplifies the difference between two input **voltages** but suppresses any voltage common to the two inputs. A differential amplifier is an analog circuit with two inputs (V_1 and V_2) and one output (V_0) in which the output is ideally proportional to the difference between the two voltages.

The formula for a simple differential amplifier can be expressed:

$$V_0 = A_d(V_1 - V_2)$$

Where

- V_0 is the output voltage
- V_1 and V_2 are the input voltages
- A_d is the gain of the amplifier (i.e. the differential amplifier gain)

From the formula above, you can see that when $V_1 = V_2$, V_0 is equal to zero, and hence the output voltage is suppressed. But any difference between inputs V_1 and V_2 is multiplied (i.e. amplified) by the differential amplifier gain A_d .

This is why the differential amplifier is also known as a difference amplifier – the difference between the input voltages is amplified.

There are two different types of differential amplifier circuits:

1. BJT Differential Amplifier – This is a differential amplifier built using transistors, either Bipolar Junction Transistors (BJTs) or Field Effect Transistors (FETs)
2. Opamp Differential amplifiers built using Operational Amplifiers

❖ Example

- 1) Design an op-amp circuit with inputs v_1 and v_2 such that $v_0 = -5v_1 + 3v_2$
- 2) Obtain v_0 for each of the circuits shown in Figure 4.9

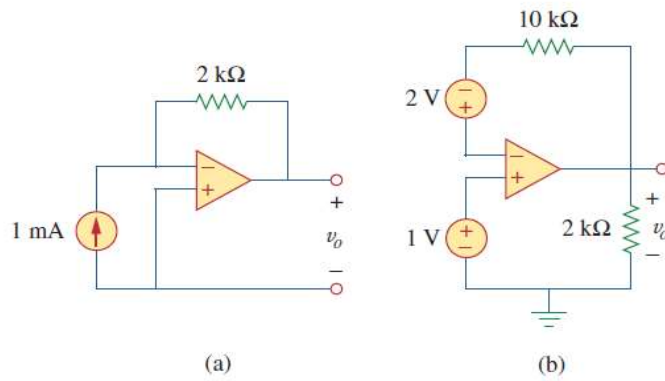


Figure 4.9 Problem 2

3) Determine the output voltage v_o in the circuit shown in Figure 4.10

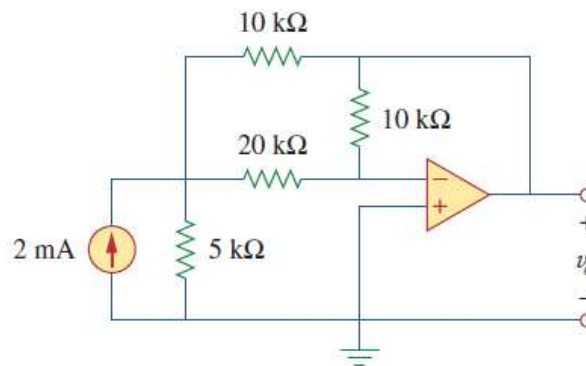


Figure 4.10 Problem 3

4) The circuit shown in Figure 4.11 is for a differential amplifier. Find v_o given that $v_1 = 1\text{ V}$ and $v_2 = 2\text{ V}$

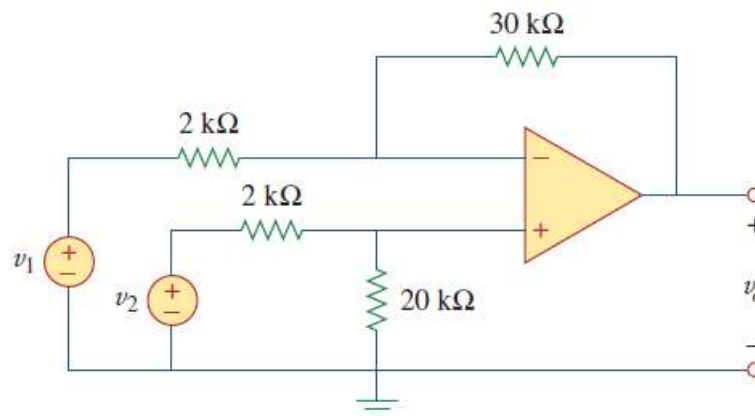


Figure 4.11 Problem 4

- 5) The circuit shown in Figure 4.12 is a differential amplifier driven by a bridge. Find v_o

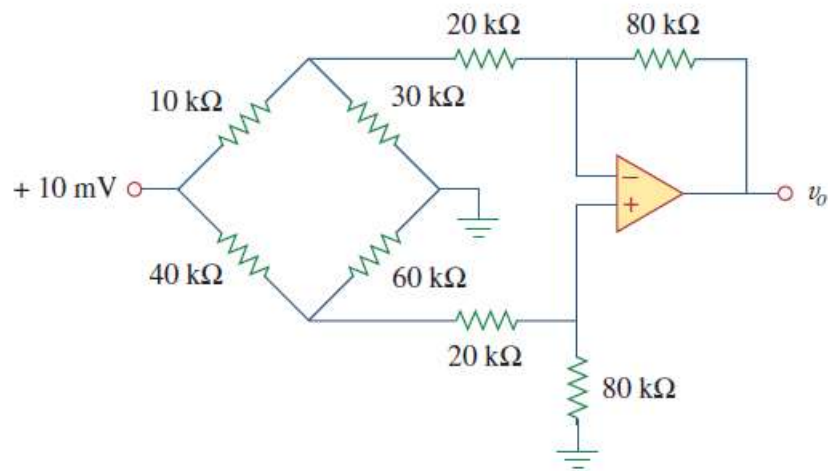


Figure 4.12 Problem 5

❖ Common-Mode Rejection Ratio (CMRR)

- When the same input voltage is applied to both input terminals of an op-amp, the op-amp is said to be operating in a common-mode configuration. Since the input voltage applied is common to both the inputs, it is referred to as a common-mode voltage v_{cm}
- A common-mode voltage v_{cm} can be ac, dc, or combination of ac and dc. Figure 4.13 shows the op-amp connected in common-mode configuration.

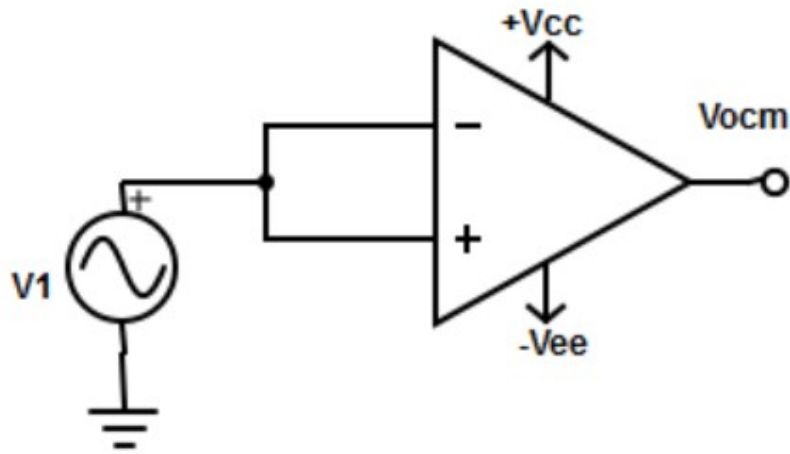


Figure 4.13 Op-amp connected in common-mode configuration

- Because ideally an op-amp amplifies only differential input voltages, no common-mode output voltage v_{ocm} should appear at the output
- However, due to imperfections within an actual op-amp, some common-mode voltage v_{ocm} will appear at the output. The amplitude of this v_{ocm} is very small and often insignificant compared to v_{cm} .
- Therefore, in practice the ratio of the v_{ocm} to the v_{cm} , which is called the common-mode voltage gain A_{cm} is generally smaller than 1. Ideally the common-mode voltage gain A_{cm} is zero.

$$A_{cm} = \frac{v_{ocm}}{v_{cm}}$$

- A_{cm} can be calculated for a given op-amp by applying a known value of common-mode input voltage v_{cm} and measuring the resultant output common-mode output voltage v_{ocm}
- Op-amp manufacturers usually list a common-mode rejection ratio (CMRR). The CMRR can be defined as the ratio of the differential gain A_D to the common-mode gain A_{cm}

$$CMRR = \frac{A_d}{A_{cm}}$$

- The CMRR can also be expressed as the ratio of the change in input offset voltage to the total change in common-mode voltage. Thus,

$$CMRR = \frac{V_{io}}{v_{cm}}$$

$$v_{ocm} = \frac{A_d v_{cm}}{CMRR}$$

These equations indicate that the higher value of CMRR, the smaller will be the amplitude of the output common-mode voltage.

- CMRR value is very large and is therefore usually expressed in decibels (dB),

$$CMRR(dB) = 20 \log \left(\frac{A_d}{A_{cm}} \right)$$

or

$$CMRR(dB) = 20 \log \left(\frac{V_{io}}{v_{cm}} \right)$$

LECTURE 3

Inverting Amplifier

An amplifier which produces a phase shift of 180° between input and output is called inverting amplifier. Fig. shows an inverting amplifier using op-amp.

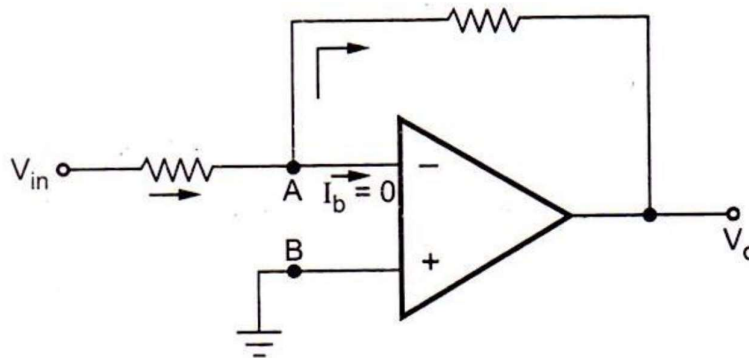


Fig. 12 Inverting amplifier

From the circuit, the potential at node B, $V_B = 0$. From the concept of virtual ground, the two input terminals are at same potential. Therefore, the potential at node A, $V_A = V_B = 0$. From the circuit,

From the circuit,

$$I_1 = \frac{V_{in} - V_A}{R_1} = \frac{V_{in} - 0}{R_1} \quad (\because V_A = 0)$$

$$I_1 = \frac{V_{in}}{R_1}$$

$$I_f = \frac{V_A - V_o}{R_f} = \frac{0 - V_o}{R_f}$$

$$I_f = -\frac{V_o}{R_f}$$

Since op-amp input current is zero, I_1 passes through R_f as I_f . That is,

$$I_1 = I_f$$

$$\frac{V_{in}}{R_1} = -\frac{V_o}{R_f}$$

$$V_o = -\left(\frac{R_f}{R_1}\right)V_{in}$$

Here R_f / R_1 is called the gain of the amplifier and negative sign indicates that the output is inverted.

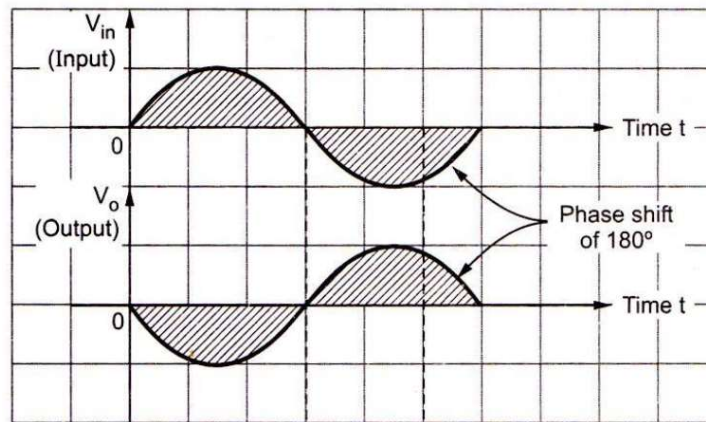


Fig. Waveforms of inverting amplifier

Non-Inverting Amplifier

An amplifier which amplifies the input without producing any phase shift between input and output is called non-inverting amplifier. Fig. shows a non-inverting amplifier using op- amp.

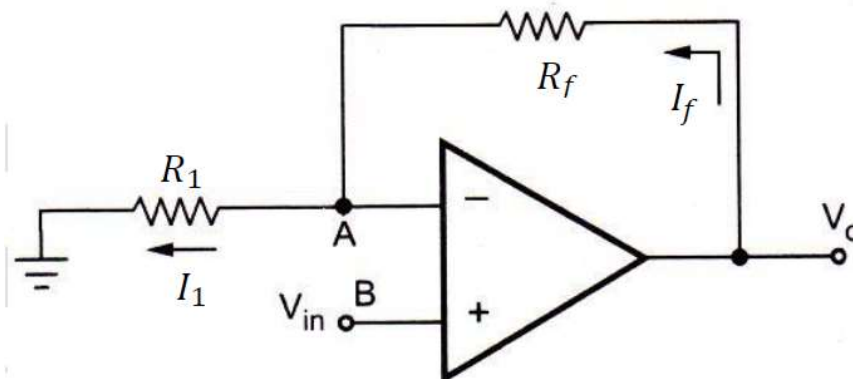


Fig. Non-inverting amplifier

From the circuit, the potential at node B, $V_B = V_{in}$. From the concept of virtual ground, the two input terminals are at same potential. Therefore, the potential at node A, $V_A = V_B = V_{in}$.

From the circuit,

$$I_1 = \frac{V_A - 0}{R_1} = \frac{V_{in} - 0}{R_1} \quad (\because V_A = V_{in})$$

$$I_1 = \frac{V_{in}}{R_1}$$

$$I_f = \frac{V_o - V_A}{R_f} = \frac{V_o - V_{in}}{R_f}$$

$$I_f = \frac{V_o - V_{in}}{R_f}$$

Since op-amp input current is zero, I_f passes through R_1 as I_1 . That is,

$$I_1 = I_f$$

$$\frac{V_{in}}{R_1} = \frac{V_o - V_{in}}{R_f}$$

$$\frac{V_{in}}{R_1} = \frac{V_o}{R_f} - \frac{V_{in}}{R_f}$$

$$\frac{V_o}{R_f} = \frac{V_{in}}{R_1} + \frac{V_{in}}{R_f}$$

$$\frac{V_o}{R_f} = \left(\frac{1}{R_1} + \frac{1}{R_f} \right) V_{in}$$

$$V_o = R_f \left(\frac{R_1 + R_f}{R_1 R_f} \right) V_{in}$$

$$V_o = \left(\frac{R_1 + R_f}{R_1} \right) V_{in}$$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) V_{in}$$

Here is called the gain of the amplifier. Fig. shows the input and output waveforms of an inverting amplifier.

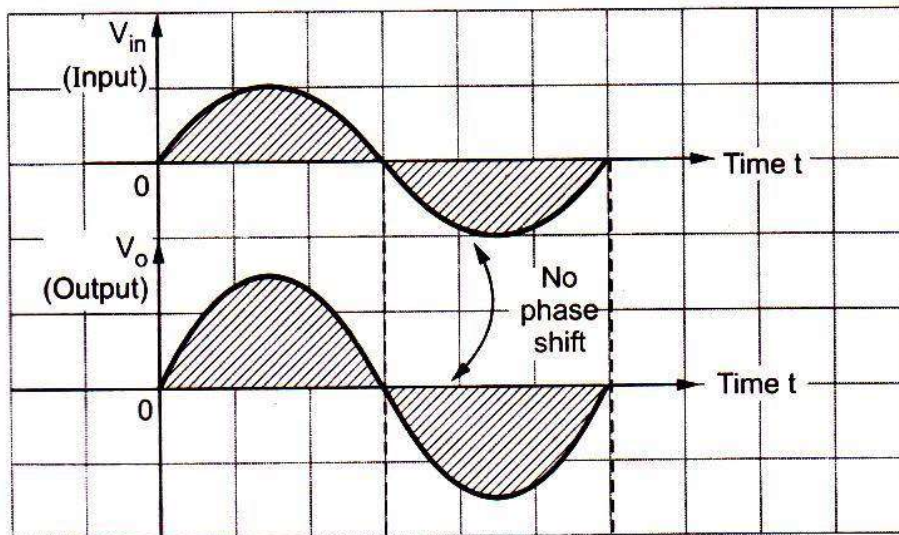


Fig. Waveforms of non-inverting amplifier

Q1) Design a circuit with one op-amp that provides a gain of 5.5. Assume you have a resistor $R_i = 10\text{k}\Omega$, what value would you choose for R_f ?

Solution

The gain here is positive hence it is a non inverting amplifier.

$$V_{in} = \frac{R_i}{R_i + R_f} \times V_{out}$$

Ideal summing point: $V_1 = V_{in}$

Voltage gain A is equal to: $\frac{V_{out}}{V_{in}}$

$$\text{Then, } A = \frac{V_{out}}{V_{in}} = \frac{R_i + R_f}{R_i}$$

$$\text{Transpose to give, } A = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_i}$$

$$5.5 = 1 + \frac{R_f}{10}$$

$$\text{Therefore } R_f = 45\text{ K}\Omega$$

Q2) Determine the input impedance and output voltage for the circuit in Figure

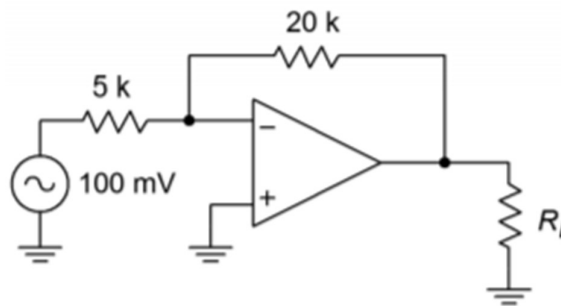


Figure: Inverting amplifier

The input impedance is set by $R_i = 5\text{k}\Omega$, therefore $Z_{in} = 5\text{k}\Omega$.

$$V_{out} = V_{in} A_v$$

$$A_v = -\frac{R_f}{R_i}$$

$$A_v = -\frac{20k}{5k}$$

$$A_v = -4$$

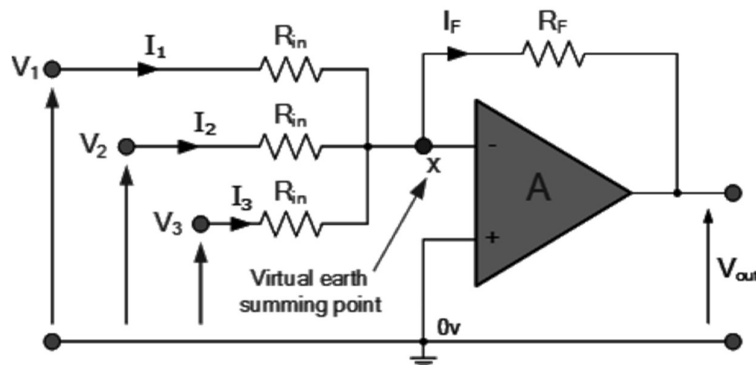
$$V_{out} = 100\text{mV} \times (-4)$$

$$V_{out} = -400\text{mV, (i.e., inverted)}$$

Lecture 4

Summing Amplifier:

The inverting amplifier has a single input voltage, (V_{in}) applied to the inverting input terminal. If we add more input resistors to the input, each equal in value to the original input resistor, (R_{in}) we end up with another operational amplifier circuit called a **Summing Amplifier**, “*summing inverter*” or even a “*voltage adder*” circuit as shown below.



In this simple summing amplifier circuit, the output voltage, (V_{out}) now becomes proportional to the sum of the input voltages, V_1 , V_2 , V_3 , etc. Then we can modify the original equation for the inverting amplifier to take account of these new inputs thus:

$$I_F = I_1 + I_2 + I_3 = - \left[\frac{V_1}{R_{in}} + \frac{V_2}{R_{in}} + \frac{V_3}{R_{in}} \right]$$

$$\text{Inverting Equation: } V_{out} = - \frac{R_f}{R_{in}} \times V_{in}$$

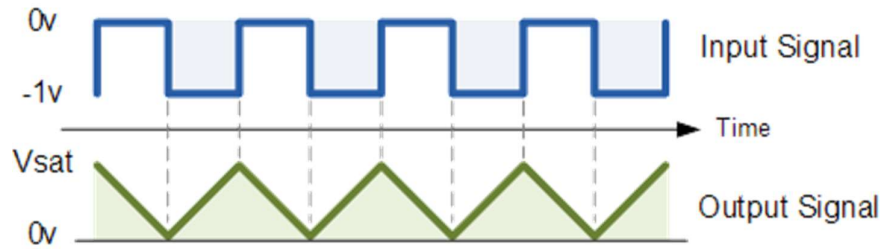
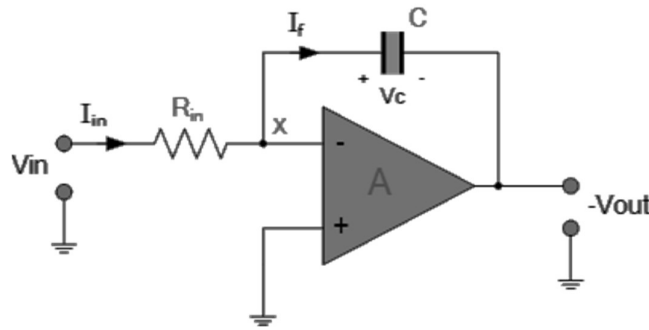
$$\text{then, } -V_{out} = \left[\frac{R_F}{R_{in}} V_1 + \frac{R_F}{R_{in}} V_2 + \frac{R_F}{R_{in}} V_3 \right]$$

A **Scaling Summing Amplifier** can be made if the individual input resistors are “NOT” equal. Then the equation would have to be modified to:

$$-V_{OUT} = V_1 \left(\frac{R_f}{R_1} \right) + V_2 \left(\frac{R_f}{R_2} \right) + V_3 \left(\frac{R_f}{R_3} \right) \dots \text{etc}$$

Op-amp Integrator Circuit:

Op-amp Integrator is an operational amplifier circuit that performs the mathematical operation of **Integration**, that is we can cause the output to respond to changes in the input voltage over time as the op-amp integrator produces an *output voltage which is proportional to the integral of the input voltage*.



We know from first principals that the voltage on the plates of a capacitor is equal to the charge on the capacitor divided by its capacitance giving Q/C . Then the voltage across the capacitor is output V_{out} therefore: $-V_{out} = Q/C$. If the capacitor is charging and discharging, the rate of change of voltage across the capacitor is given as:

$$V_c = \frac{Q}{C}, \quad V_c = V_x - V_{out} = 0 - V_{out}$$

$$\therefore -\frac{dV_{out}}{dt} = \frac{dQ}{Cdt} = \frac{1}{C} \frac{dQ}{dt}$$

But dQ/dt is electric current and since the node voltage of the integrating op-amp at its inverting input terminal is zero, $X = 0$, the input current I_{in} flowing through the input resistor, R_{in} is given as:

$$I_{in} = \frac{V_{in} - 0}{R_{in}} = \frac{V_{in}}{R_{in}}$$

The current flowing through the feedback capacitor C is given as:

$$I_f = C \frac{dV_{out}}{dt} = C \frac{dQ}{Cdt} = \frac{dQ}{dt} = \frac{dV_{out} \cdot C}{dt}$$

Assuming that the input impedance of the op-amp is infinite (ideal op-amp), no current flows into the op-amp terminal. Therefore, the nodal equation at the inverting input terminal is given as:

$$I_{in} = I_f = \frac{V_{in}}{R_{in}} = \frac{dV_{out} \cdot C}{dt}$$

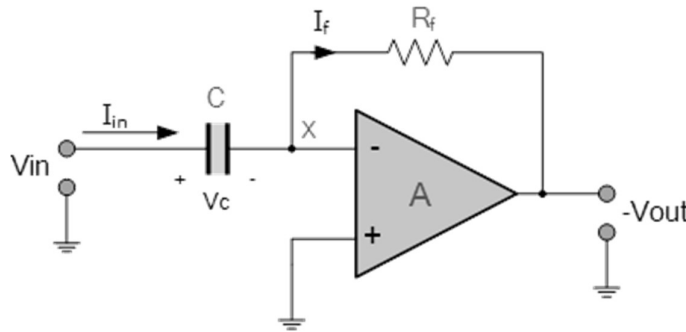
$$\therefore \frac{V_{in}}{V_{out}} \times \frac{dt}{R_{in} C} = 1$$

From which we derive an ideal voltage output for the **Op-amp Integrator** as:

$$V_{out} = -\frac{1}{R_{in} C} \int_0^t V_{in} dt = -\int_0^t V_{in} \frac{dt}{R_{in} \cdot C}$$

Op-amp Differentiator Circuit:

Here, the position of the capacitor and resistor have been reversed and now the reactance, X_C is connected to the input terminal of the inverting amplifier while the resistor, R_f forms the negative feedback element across the operational amplifier as normal. This operational amplifier circuit performs the mathematical operation of **Differentiation**, that is it “*produces a voltage output which is directly proportional to the input voltage’s rate-of-change with respect to time*”.



The input signal to the differentiator is applied to the capacitor. The capacitor blocks any DC content so there is no current flow to the amplifier summing point, X resulting in zero output voltage. The capacitor only allows AC type input voltage changes to pass through and whose frequency is dependant on the rate of change of the input signal. Since the node voltage of the operational amplifier at its inverting input terminal is zero, the current, i flowing through the capacitor will be given as:

$$I_{IN} = I_F \text{ and } I_F = -\frac{V_{OUT}}{R_F}$$

The charge on the capacitor equals Capacitance times Voltage across the capacitor:

$$Q = C \times V_{IN}$$

Thus, the rate of change of this charge is:

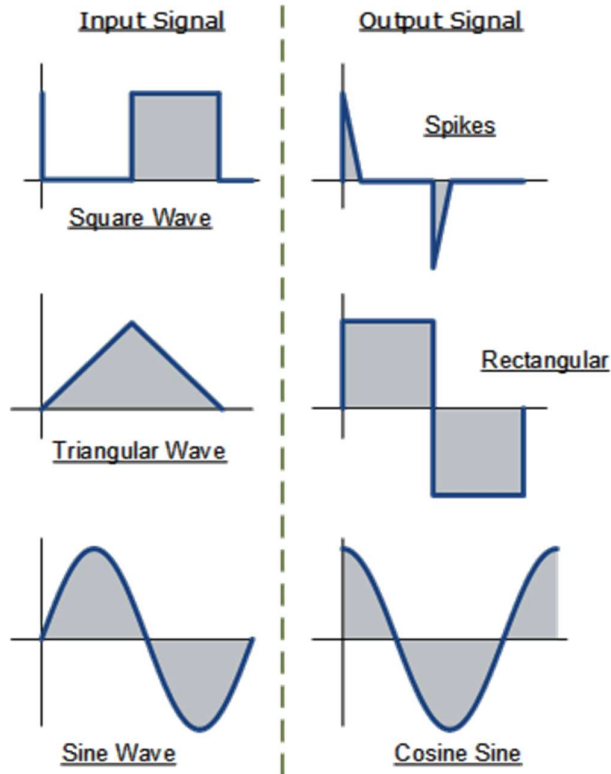
$$\frac{dQ}{dt} = C \frac{dV_{IN}}{dt}$$

But, dQ/dt is the capacitor current, i

$$I_{IN} = C \frac{dV_{IN}}{dt} = I_F$$
$$\therefore -\frac{V_{OUT}}{R_F} = C \frac{dV_{IN}}{dt}$$

From which we have an ideal voltage output for the op-amp differentiator is given as:

$$V_{OUT} = -R_F C \frac{dV_{IN}}{dt}$$



Exercise Problems

1. Calculate the output voltage for the circuit of Fig. 4.1. The inputs are $V_1 = 50 \text{ mV} \sin(1000t)$ and $V_2 = 10 \text{ mV} \sin(3000t)$.

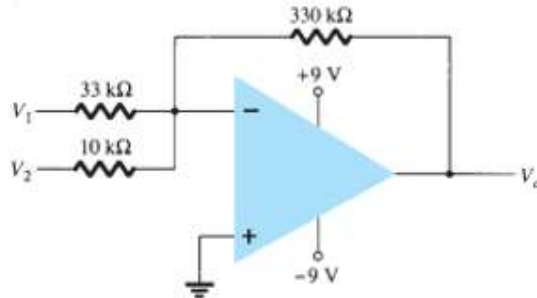


Fig. 4.1

2. Calculate the output voltage (V_{out}) for the circuit shown in Fig. 4.2.

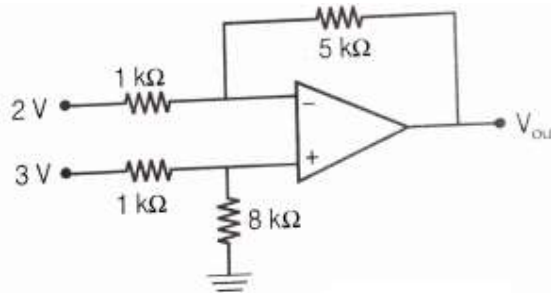


Fig. 4.2

3. The inverting OP-AMP shown in Fig. 4.3 has an open loop gain of 100. Calculate closed loop gain of the circuit.

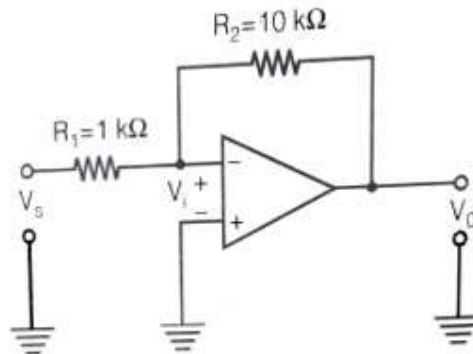


Fig. 4.3

4. Assume that the inverting OP-AMP shown in Fig. 4.4 is ideal. Find the value of current flowing through the $1 \text{ K} \Omega$ resistor.

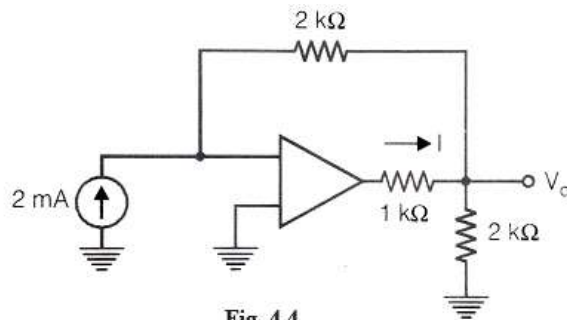


Fig. 4.4

5. Assume that the OP-AMP shown in Fig. 4.5 is ideal. Calculate the value of R in kΩ, If the voltage inputs are to be amplified by the same amplification factor.

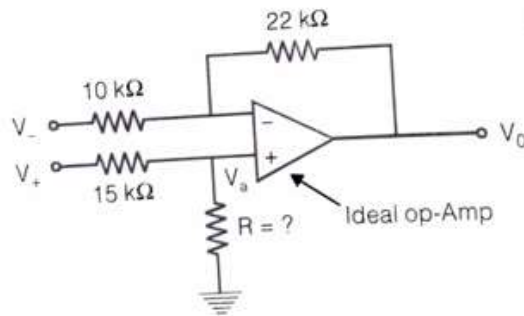


Fig. 4.5

Solution:

1. Calculate the output voltage for the circuit of Fig. 4.1. The inputs are $V_1 = 50 \text{ mV} \sin(1000t)$ and $V_2 = 10 \text{ mV} \sin(3000t)$.

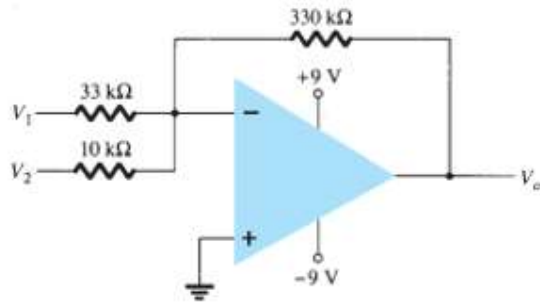


Fig. 4.1

Solution

The output voltage is

$$\begin{aligned} V_o &= -\left(\frac{330 \text{ k}\Omega}{33 \text{ k}\Omega} V_1 + \frac{330 \text{ k}\Omega}{10 \text{ k}\Omega} V_2\right) = -(10 V_1 + 33 V_2) \\ &= -[10(50 \text{ mV}) \sin(1000t) + 33(10 \text{ mV}) \sin(3000t)] \\ &= -[0.5 \sin(1000t) + 0.33 \sin(3000t)] \end{aligned}$$

2. Calculate the output voltage (V_{out}) for the circuit shown in Fig. 4.2.

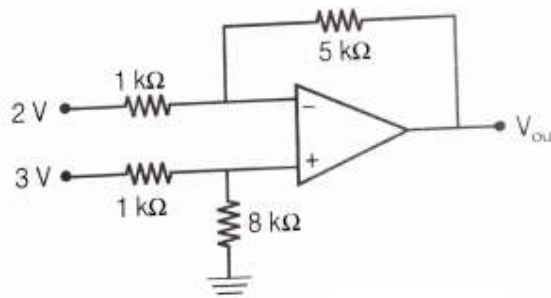


Fig. 4.2

Solution:

$$\begin{aligned} V_o &= -\frac{R_F}{R_1} \times 2 + \left(1 + \frac{R_F}{R_1}\right) \cdot \frac{8}{9} \times 3 \\ &= -5 \times 2 + 6 \times \frac{8}{3} \\ V_o &= 6V \end{aligned}$$

3. The inverting OP-AMP shown in Fig. 4.3 has an open loop gain of 100. Calculate closed loop gain of the circuit.

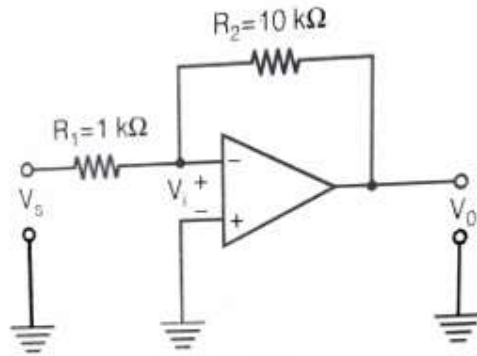


Fig. 4.3

Solution:

If open loop gain is not infinite, then

$$\frac{1}{\beta} = -\frac{R_f}{R_i} = -\frac{10k}{1k} = -10$$

$$|\beta| = \frac{1}{10}$$

$$A_f = \frac{-A}{1 + A\beta} = \frac{-100}{1 + 100 \times \frac{1}{10}}$$

$$A_f \approx -9$$

4. Assume that the inverting OP-AMP shown in Fig. 4.4 is ideal. Find the value of current flowing through the $1K \Omega$ resistor.

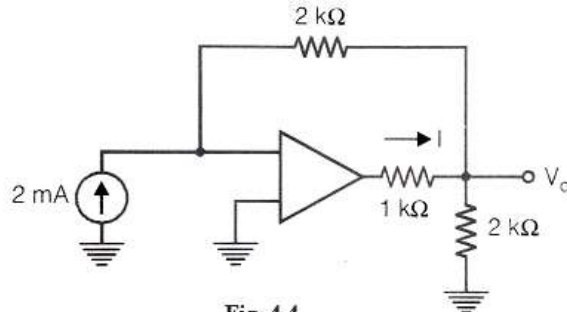


Fig. 4.4

Solution:

$$V_o = -2 \times 10^{-3} \times 2 \times 10^3 = -4 \text{ V}$$

Apply KCL at output node

$$I + 2 \text{ mA} = \frac{V_o}{2k} = -\frac{4}{2k} = -2 \text{ mA}$$

$$I = -4 \text{ mA}$$

5. Assume that the OP-AMP shown in Fig. 4.5 is ideal. Calculate the value of R in $k\Omega$, If the voltage inputs are to be amplified by the same amplification factor.

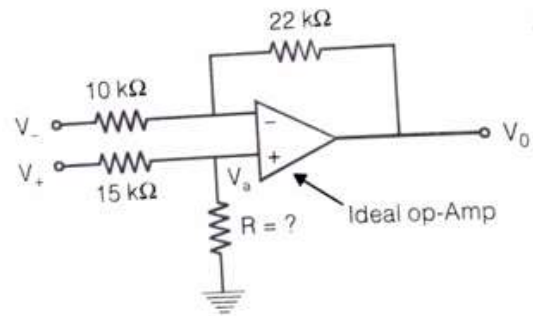


Fig. 4.5

Solution:

$$A^- = -\frac{R_f}{R_1}$$

$$A^- = -\frac{22k}{10k} = -2.2$$

$$|A^-| = 2.2$$

$$V_a = V_+ \left(\frac{R}{R+15} \right)$$

$$A^+ = 1 + \frac{R_f}{R_1}$$

$$\frac{V_o}{V_a} = 1 + \frac{R_f}{R_1} = \frac{V_o}{V_+ \left(\frac{R}{R+15} \right)}$$

$$\frac{V_o}{V_+} = \left(1 + \frac{22}{10} \right) \times \left(\frac{R}{R+15} \right)$$

$$|A^+| = 3.2 \left(\frac{R}{R+15} \right)$$

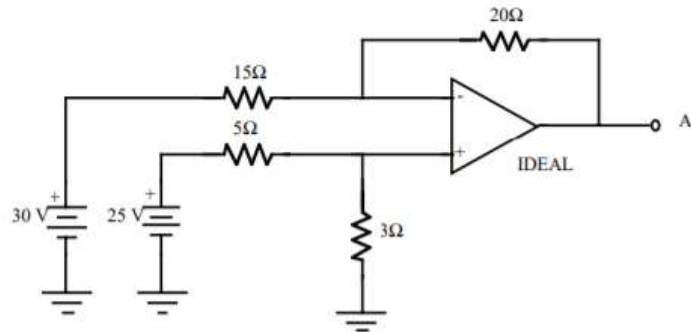
$$\frac{3.2R}{R+15} = 2.2$$

$$3.2R = 2.2R + 33$$

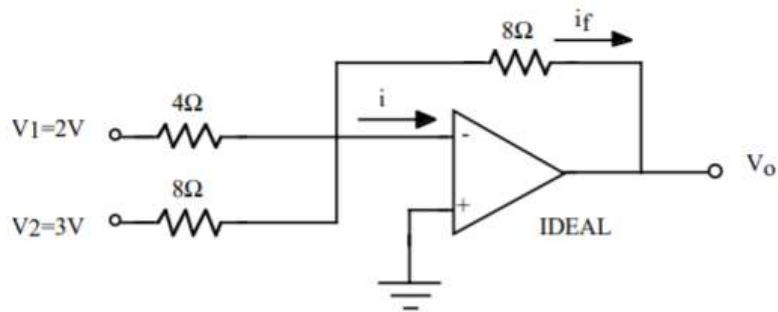
$$R = 33 \text{ k}\Omega$$

MCQ

1. For the difference amplifier circuit shown in following circuit, determine the output voltage at terminal A.

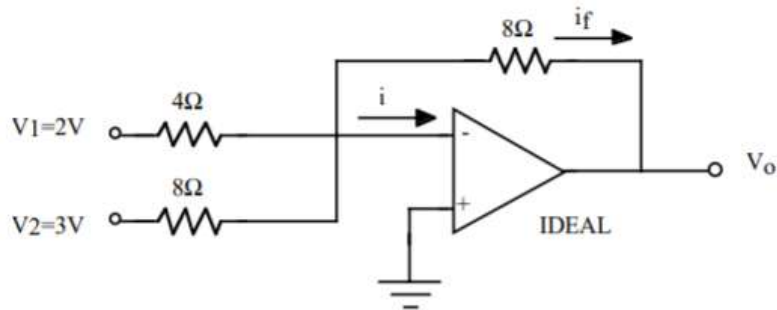


- (a) - 18.13 V
 (b) -6.07 V
 (c) 6.07 V
 (d) 15.45 V
2. What is the current 'i' in following circuit?



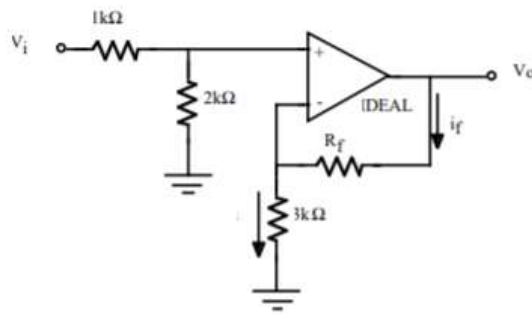
- (a) 0.88 A
 (b) 0.25 A
 (c) 0
 (d) -0.25 A

3. What is the output voltage V_o ?



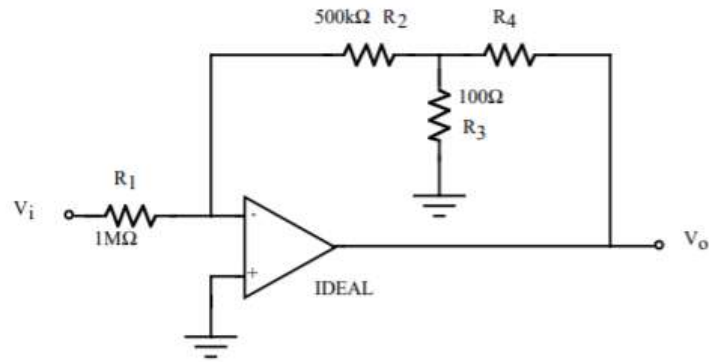
- (a) -7 V
- (b) -6 V
- (c) -1 V
- (d) 6 V

4. For the ideal op amp shown, what should be the value of resistor R_f to obtain a gain of 5?



- (a) $12.0\text{ k}\Omega$
- (b) $19.5\text{ k}\Omega$
- (c) $22.5\text{ k}\Omega$
- (d) $27.0\text{ k}\Omega$

5. Evaluate the following amplifier circuit to determine the value of resistor R_4 in order to obtain a voltage gain of -120 .



- (a) 12 kΩ
- (b) 19 kΩ
- (c) 24 kΩ
- (d) 25 kΩ

Solutions

Sol. 1.

By voltage division,

$$v_{in+} = 25V \left(\frac{3\Omega}{5\Omega + 3\Omega} \right) = 9.375V$$

By the virtual short circuit between the input terminals, $v_{in-} = 9.375V$

Using **Ohm's** law, the current through the 15 Ω resistor is

$$I_{15} = \left(\frac{30V - 9.375V}{15\Omega} \right) = 1.375V$$

The input impedance is infinite; therefore, $I_{in-} = 0$ and $I_{15} = I_{20}$.

Use Kirchoff's voltage law to find the output voltage at A.

$$v_A = v_{in-} - 20I_{20} = 9.375V - (20\Omega)(1.375A) = -18.125V$$

Answer is A.

Sol. 2

The input current in an op amp is so small that it is assumed to be zero.

Answer is C.

Sol. 3

This op amp circuit is a summing amplifier. Since $i=0$,

$$i_f = \frac{v_1}{R_1} + \frac{v_2}{R_2} = \frac{3 \text{ V}}{8 \Omega} + \frac{2 \text{ V}}{4 \Omega} = 0.875 \text{ A}$$

$$v_o = -i_f R_f = -(0.875 \text{ A})(8 \Omega) = -7 \text{ V}$$

Answer is A.

Sol. 4

By voltage division, $v_{in+} = v_i \left(\frac{2k\Omega}{3k\Omega} \right) = \frac{2}{3} v_i$

By the virtual short circuit, $v_{in-} = v_{in+} = \frac{2}{3} v_i$

$$i = \frac{v_{in-}}{3k\Omega} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

Since the op amp draws no current, $i_f = i$

$$\frac{v_o - v_{in-}}{R_f} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

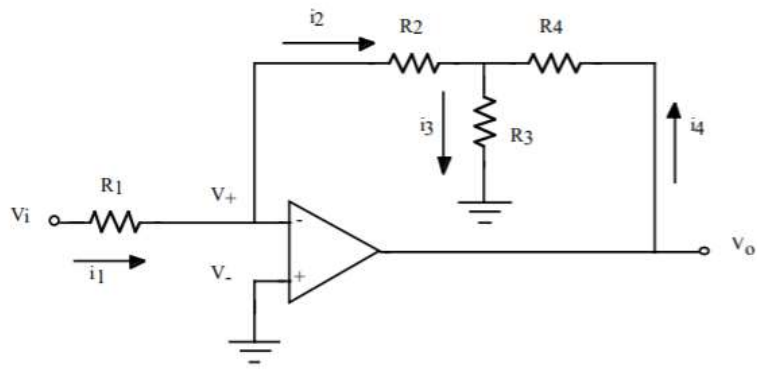
But, $v_o = 5v_i$.

$$\frac{5v_i - \frac{2}{3} v_i}{R_f} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

$$\frac{\frac{13}{3}}{R_f} = \frac{\frac{2}{3}}{3k\Omega}$$

$R_f = 19.5 \text{ k}\Omega$, Answer is B.

Sol. 5



v_{in+} is grounded, so v_{in-} is also a virtual ground.

$$v_{in-} = 0$$

Since $v_{in-} = 0$, $v_i = i_1 R_1$ and $i_1 = v_i / R_1$.

Since $v_{in-} = 0$, $v_x = -i_2 R_2$ and $i_2 = -v_x / R_2$.

Similarly,

$$v_x = -i_3 R_3$$

$$v_x - v_o = -i_4 R_4$$

From Kirchhoff's current law,

$$i_4 = i_2 + i_3$$

$$\frac{v_x - v_o}{R_4} = \frac{-v_x}{R_2} + \frac{-v_x}{R_3}$$

Now, $v_o = -120v_i$.

Also, $i_1 = i_2$, so

$$\frac{v_i}{R_1} = \frac{-v_x}{R_2}$$

$$v_x = -\left(\frac{R_2}{R_1}\right)v_i$$

$$\frac{-\left(\frac{R_2}{R_1}\right)v_i - (-120v_i)}{R_4} = \frac{\left(\frac{R_2}{R_1}\right)v_i}{R_2} + \frac{\left(\frac{R_2}{R_1}\right)v_i}{R_3}$$

$$\frac{120\left(\frac{R_1}{R_2}\right) - 1}{R_4} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{R_2 + R_3}{R_2 R_3}$$

$$R_4 = \frac{120\left(\frac{R_1}{R_2}\right) - 1}{\frac{R_2 + R_3}{R_2 R_3}} = \frac{120\left(\frac{1 \times 10^6 \Omega}{5 \times 10^5 \Omega}\right) - 1}{\frac{5 \times 10^5 \Omega + 100 \Omega}{(5 \times 10^5 \Omega)(100 \Omega)}} = 2.39 \times 10^4 \Omega \text{ (24 k}\Omega\text{)}$$

Answer is C.